Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/10/2011
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; STS-135		
Key Personnel Changes/Previous PI:	Viju Gupta, PhD has been promoted from Post-Doc to a Research Assistant Professor, but remains as a key personnel on this grant.		
COI Name (Institution):	Roby, Katherine (University of Kansas Medical Center) Gupta, Vijayalaxmi (University of Kansas Medical Center)		
Grant/Contract No.:	NNX09AP04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Published data demonstrates that short term microgravity (µG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based µG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to µG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term µG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to µG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30d of µG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term µG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to µG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process.
Task Progress:	Microgravity broadly degrades physiological systems including bone, immune response, muscle, cardiovascular, wound healing, and vestibular systems. Many of these changes also occur as part of the natural aging process, but are greatly accelerated in space flight. Emerging data shows that all of these systems are modulated by estrogen receptor signaling. This suggests the question: Does space flight alter ovarian function? No flight data for normal adult estrous cycling female rodents has been published. No detailed human data has been published on effects of space flight on normal ovarian function. However, female astronauts are recommended to take contraceptives during space flight to prevent cycling and pregnancy; and there is data on post-flight difficulties getting pregnant. Our flight data from STS-131 demonstrate that Micro gravity negatively impacts ovarian histology in mice; manifested by lower numbers of corpora lutea, atretic (dying) follicles, and unhealthy oocytes. Estrogen receptor levels were found to be lower in flight mice. The combined data suggest that oocyte maturation and production were blocked or terminated after 15 days of space flight. Since ovaries have a limited population of maturation competent follicles, and atretic follicles do not recover, space flight may shorten reproductive life span. Preliminary analysis of our data from both STS-133 and STS-135 confirms the results obtained from STS-131. For STS-133, even though the untreated n's were lower than for STS-131 and STS-135, the flight ovaries for STS-133 still showed a highly significant decline in corporal lutea (to 0, $P<0.02$). Taken together, our mouse data suggest that decrements in gonadal endocrine function and steroid hormone receptor signaling may be an underlying mechanism in space flight induced changes on wider physiological systems. Our research in both flight and ground-based systems is exploring the mechanisms that involve these space-related changes, how they relate to the normal aging process, and common mechanisms that could be used to develop countermeasures the will ameliorate both space flight effects on these systems, but also provide solutions to improve the lives and reduce the health risks of the aging individual on earth.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Tash JS, Gupta V, Holets L, Roby KF. "STS-131: Spaceflight has negative impacts on the morphology, follicle health, and steroid hormone receptors in ovaries and uterine horns in C57Bl/6J mice." Presented at the 18th IAA Humans in Space Symposium, Houston TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston TX, April 11-15, 2011. , Apr-2011
Significant Media Coverage	Bavley A. "Front page article: Sex and Space? Houston, we have a problem." The Kansas City Star, January 5, 2011., Jan-2011
Significant Media Coverage	McKean ML. "Mice on the Shuttle Are Part of KU Med Ctr Research. TV interview about Tash experiment on STS-131 & STS-135." TV interview: WDAF Kansas City, July 2011. http://www.fox4kc.com/health/wdaf-ku-researchers-wait-for-space-mice-return-20110718 ,0,5216433.story ; Accessed 8/10/2011., Jul-2011